

Exam 1 answers

Question 1:

The correct answer is: **21 → 16 → 18 → 17 → 10 → 3.**

We will also accept as correct: **21 → 16 → 10 → 3**

Explanation: We first compute the derivative of the image in the y direction. The only pixels whose derivatives are > 6 are the 10 pixels that make up the base of each triangle.

Therefore, taking a threshold at 6, summing all pixels and dividing by 10 will give us the number of triangles.

Note: In this case, blurring the image first is wrong -- this will "smear" the edges, and therefore mess up the counting by dividing by precisely 10.

Question 2:

One correct answer is: **1 → 2 → 15 → 18** (based on histogram equalization, and the understanding that a threshold at $M=26$ is $\sim 10\%$ of the gray levels of the image).

Another correct answer: **17 → 9 → 25 → 8 → 11 → 24 → 4 → 27 → 18** (based on quantization of the original gray levels into two values, where the algorithm iterates until the q 's (means) of each value do not change anymore).